THE ATOMIC THEORY

Development of modern atomic theory was based on two types of research carried out by many scientists just before and after 1900. The first type of research was based on electrical nature of matter and this led to the discovery of the compositions of the atom. The second type of research was centred on the interaction of matter with energy—this led to a much more detailed understanding of the arrangement of particles in atoms. The arrangement of particles in atoms determines the chemical and physical properties of each element.

Dalton's Atomic Theory

Joseph Proust (1754–1826) performed series of experiments to determine the relative masses of different elements in compounds. He discovered that the ratio of the masses of elements was always the same. Therefore, he proposed the law of constant composition (law of definite proportions), which states that all samples of a given compound have the same mass ratio of their constituent. However, Proust could not vividly explain the law of definite proportions using ideas of individual atom. His work led to search for understanding of the atomic nature of matter.

John Dalton (an English schoolteacher and a chemist) made the first convincing argument for atoms in 1808. John Dalton built on Proust's work by investigating the mass ratios in groups of compounds consisting of the same elements. He measured the ratios of the masses of elements that can combine to form compounds and found that the ratios formed patterns e.g. there are 16 g of oxygen for every 1 g of hydrogen. Dalton's work led to the first tabulation of relative weights of elements and his atomic theory was key in explaining the atomic nature of matter.

This is the summary of Dalton's atomic theory:

1. all chemical elements are composed of indivisible particles called atoms;
2. atoms cannot be created nor destroyed (law of conservation of mass). Chemical reactions only rearrange the manner in which atoms are combined;
3. all the atoms of one elements are identical and different from those of other elements (law of constant proportion);
4. atoms combine to form compounds in the ratio of small whole numbers (law of multiple proportion).

Modern instrumentation provides direct evidence of atoms. Atoms exist and they are the units that make up elements. An element is composed of only one type of atom. All matters are made up of various combinations of the simple forms of matter, known as the chemical elements.

The Dalton's atomic theory has been modified because of new findings. Atoms can be subdivided or destroyed. Atoms of an element maybe different. Isotopes are atoms of the same element that are different in masses. Some atoms, such as carbon, that form rings and long chains compounds can form giant molecules containing thousands of atoms. This indicates that combinations of atoms to form compounds may not be in the ratio of small whole numbers.

The Fundamental Particles

Atoms consist of three fundamental particles: electron, proton, and neutron. A good nature of the knowledge of the nature and functions of these particles is necessary for understanding of chemical reactions. The properties of the three fundamental particles are shown in **Table 1.1**. The charge on an electron is equal in magnitude, but opposite in sign, to the charge on a proton. Many other subatomic particles such as quarks, positrons, mesons, antiprotons, gluons, pions, muons and neutrinos have also been discovered.

Table 1.1: properties of fundamental particles

Particle	**Symbol**	**Mass (amu)**	**Charge**	**Rest mass (Kg)**
Electron	e^-	0.00054858	–1	9.109×10^{-31}
Proton	P or P^+	1.0073	+1	1.673×10^{-27}
Neutron	n or n°	1.0087	None	1.673×10^{-27}

Discovery of Electrons

Humphrey Davy proposed some of the earliest evidences about the atomic structure in the early 1800. He passed electric current through some chemical substances, and the substances decomposed. Therefore, Davy proposed that the elements of a compound are held together by electrical forces. Michael Faraday, who was Davy's student, between 1832 and 1833, quantitatively determined the relationship between the amount of chemical reaction that takes place and the amount of electricity used during electrolysis. George Stoney (1826–1911) studied the work of Faraday and suggested in 1874 that units of electric charge are associated with atoms. He suggested in 1891 that the electrical charges be called electrons.

The first important experiments that led to the proper understanding of the compositions of the atom were investigated by an English physicist, Joseph John Thomson (1856–1940). He studied, between 1898 and 1903, the electrical discharges in partially evacuated tubes called *cathode-ray tubes* (**Figure 1.1**). At low pressures and high voltages, he was able to produce cathode rays that produced a well-defined spot on a fluorescent screen. The spot could be deflected by an electric field produced by secondary electrodes and by a magnetic field perpendicular to the electric field. The ray was produced at the negative electrode (cathode) and was repelled by the negative pole of an applied electric field; J.J. Thomson therefore postulated that the ray was a stream of negatively charged particles. In 1897, J.J. Thomson studied these negatively charged particles more carefully and closely and called them *electrons*, as Stoney had suggested. He studied the degree of deflections of cathode rays in different electric and magnetic fields, and used his measurement to determine the charge-to-mass ratio (e/m) of the particles forming the cathode rays.

$$\frac{e}{m} = 1.75882 \times 10^{8} \text{ C/g}$$

Thomson found that the value of e/m remained the same irrespective of the metal used for the cathode or type of gas used to fill the discharge tube or composition of the electrodes or nature of the electric powers source.

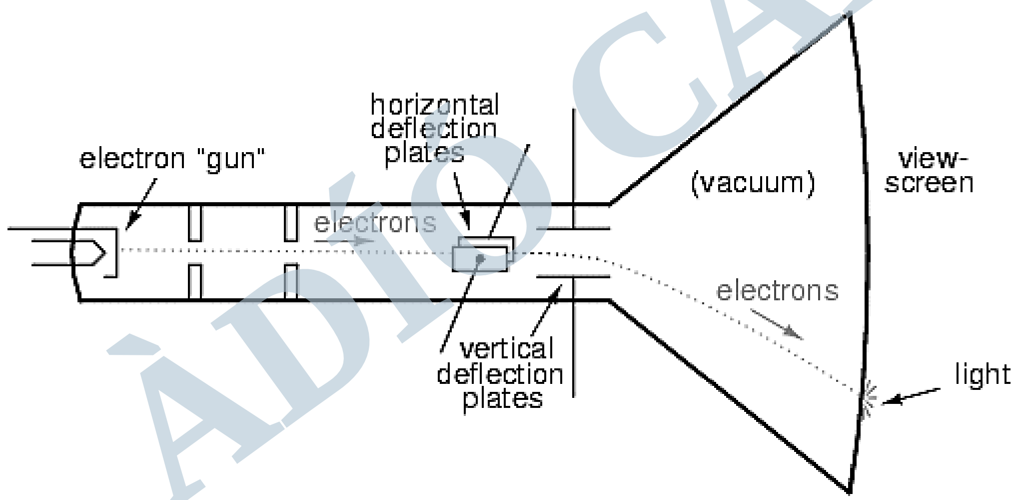


Figure 1.1*: cathode ray tube*

In a very high vacuum, cathode rays could not be detected. Cathode rays came from inside the atoms that made up the electrode called the cathode. One of the outcomes and breakthroughs of the Thomson's work was the invention of the cathode ray tube used in television.

Some of the characteristics of cathode rays are:

1. they are streams of electrons

2. they travel in a straight line from cathode to anode
3. cathode rays cast a sharp shadow of object placed on their path
4. they are deflected by both magnetic and electric fields in the direction expected for negatively charged particles
5. cathode rays produced X-rays when strike a metallic object
6. they have mass because they can turn a small paddle placed in their path
7. they cause glass to fluorescence with green light.

Robert Millikan in 1909 began another experiment that led to the accurate and direct evaluation of electrons. He determined the charge of electron in his famous "oil-drop" experiment. He observed the motion of a charged non-evaporating oil. He assumed that this smallest charge was the charge on one electron. The value of e^- is 1.602177×10^{-19} C.

Combining the value of e/m from Thomson experiment and the value of e^- from Millikan experiment, it was possible to evaluate the mass of electron. The value of mass of electron is 9.109340×10^{-28} g. The value of the mass of electron is about $\frac{1}{1836}$ the mass of hydrogen atom (the lightest atom). This is the first experiment that suggests that atoms contain an integral number of electrons.

Canal Rays and the Proton

The proton was the second sub-atomic particle to be determined. Eugen Goldstein in 1886 first observed positively charged particles emanating from perforated cathode ray tube. The positively charged particles moved towards the cathode. These were called canal rays because they were observed to pass occasionally through a channel (canal or hole) drilled in the negative cathode. When gaseous atom is placed in the tube (**Figure 1.2**), the gaseous atom will lose electron(s) and positive rays will be formed.

$$Atom \rightarrow cation + e^-$$

It was observed that different elements give positive ions with different ratios and that there exist a unit of positive charge, called proton. The proton is a fundamental particle with a charge equal in magnitude but opposite in sign to the charge on the electron. The mass of proton is almost 1836 times the mass of electron.

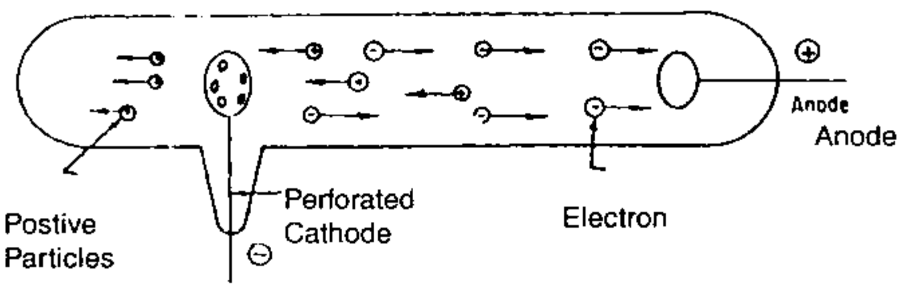


Figure 1.2*: production canal or positive rays*

Some of the characteristics of positive rays are:

1. positive rays travel in a straight line in a direction opposite to that of the cathode
2. they cause ZnS to fluorescence
3. the value of e/m of positive particles or rays depends on the nature of the gas in the tube
4. the mass is far more than the mass of electron
5. they are deflected by both electric and magnetic fields in way indicating the direction of positively charged particles.

Ernest Rutherford and the Nuclear Atom

By the early 1900, It was evident that each atom contains regions of both positive and negative charges. However, the location and arrangement of these particles were unknown. Thomson assumed that positive charge was evenly distributed throughout the atom.

Ernest Rutherford in 1909 established that α-particles are positively charged. Rutherford in his “alpha particle scattering” or “gold foil” experiment (obtained from Geiger and Marsden experiment), directed a stream of very highly and energetic α-particles from a radioactive source against a thin gold foil provided with a circular fluorescent ZnS screen. The schematic diagram that represent Rutherford experiment is shown in **Figure 1.3**. Scintillations or flashes were observed on the screen caused by individual α-particle when hit the screen. Rutherford discovered several radioactive elements and identified α- and ß-radiations.

The α-particles are extremely dense, denser than gold foil. The analysis of Rutherford results showed that the scattering was caused by repulsion from very dense regions of positive charge in the gold foil. The experiment provided evidence that the atom consisted of a very small positively charged nucleus surrounded by a large space containing light, negatively charged electrons. Rutherford then proposed the existence of the proton and proved that its mass was over 1800 times that of the electron. He determined the magnitude of the charges on the atomic nuclei. He concluded that "the atoms consist of very small, very dense positively charged nuclei surrounded by clouds of electrons at relatively distance from the nuclei". The positive charges of the atom are located on the nucleus.

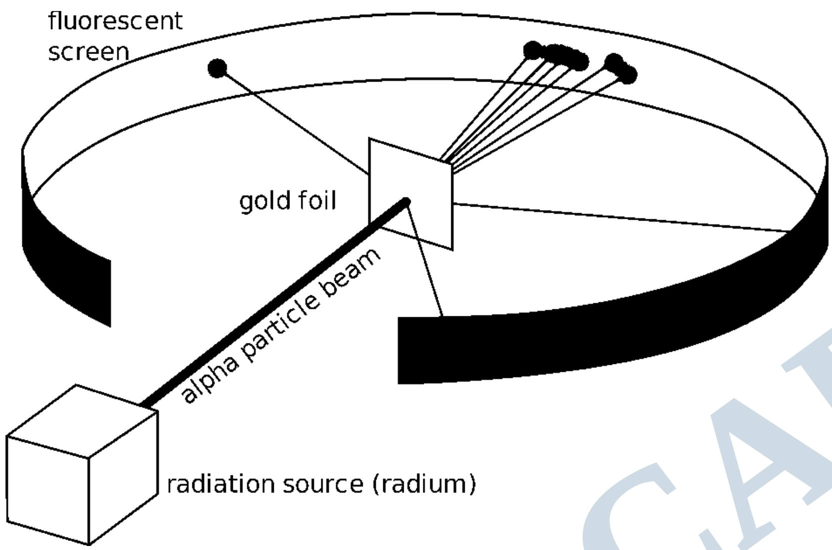


Figure 1.3*: Rutherford experiments*

The Neutron

The third fundamental particle, was discovered experimentally in 1932 by Sir James Chadwick. He bombarded Be with high-energy α-particles and correctly interpreted the results. Since these particles were neutral, they were called *neutrons*. The neutron is an uncharged particle with a mass slightly greater than that of the proton.

Atom consists of very small, very dense positively charged nuclei surrounded by clouds of light negatively charged electrons at relatively great distances from the nuclei. All nuclei contain proton; nuclei of all atoms except the common form of hydrogen ($^{1}_{1}H$) also contain neutrons. Electron could gain or lose energy by jumping from one orbit to another. Bohr's model of the atom is still in use up till today.

The current model of the atom is known as the nuclear atom. The summary of the model is:

1. atoms are made up of sub-atomic particles, namely electrons, protons and neutrons

2. the protons and neutrons form a compact, central body called the nucleus of the atom
3. the electrons are distributed in space like a cloud around the nucleus.

Atomic Number

A few years after the Rutherford experiment, H.G.J. Moseley (1913) studied the X-ray given off by various elements. Moseley generated X-rays by aiming a beam of high-energy electrons at a solid target made of a single pure element. He analysed the X-rays produced by different metals using photographic techniques and obtained a straight-line graph. On the basis of his mathematical analysis of the X-rays data, he concluded that each element differs from the preceding one by having one more positive charge in its nucleus. For the first time, it was possible to arrange all the known elements in order of increasing nuclear charge. Moseley found out that the square root of the frequency of the X-rays was directly proportional to an integral number, ***Z***.

$$Z \ \alpha \ \sqrt{frequency}$$

The number, *Z*, was approximately half the value of the atomic weight. He concluded that the number was not a mere number but a fundamental property of the atom. The number, *Z*, is called the atomic number or proton number. Atomic number is therefore the number of protons in the atom of an element. It is also equal to the number of electrons in a neutral atom. Atomic number is the same for all atoms of an element. Moseley was able to predict the existence of three missing elements with atomic numbers 43, 61 and 75, which were later identified as technetium, promethium and rhenium, respectively. He further calculated the number of units of positive charge on the nuclei of several atoms.

Mass Number and Isotopes

The mass number of an atom is the sum of the number of protons and neutrons in its nucleus. It is represented by ***A***. A nucleus is regarded as nuclide; the nuclear particles, protons and neutrons, are sometimes called nucleons. Mass number is also known as atomic mass.

A = Z + N where N is the number of neutrons in the nucleus.

The mass numbers are always indicated as superscripts while the atomic numbers are indicated as subscripts on the left of the symbol of an element.

${}^{A}_{Z}E$ where *E* represents element.

${}^{12}_{6}C$ means that carbon has atomic number of 6 and mass number of 12. ${}^{14}_{6}C$ means that nuclide of carbon has atomic number of 6 and mass number of 14. All carbon nuclides have atomic number of 6; ${}^{14}_{6}C$ nuclide can be written as ${}^{14}C$ or Carbon-14.

Example 1.1: calculate the number of protons, electrons and neutrons in the following:

(i) ${}^{35}_{17}Cl$ (ii) ${}^{37}_{17}Cl$ (iii) ${}^{24}_{12}Mg^{2+}$

Solution:

(i) ${}^{35}_{17}Cl$

The atomic number is 17, therefore, there are 17 protons in the nucleus; since the nuclide is not charged (neutral), the number of protons is equal the number of electron, then there are also 17 electrons; number of neutrons = mass number – atomic number = 35 – 17 = 18

(ii) ${}^{37}_{17}Cl$

There are 17 protons in the nucleus; there are 17 electrons; number of neutrons = 37 – 17 = 20

(iii) ${}^{24}_{12}Mg^{2+}$

The atomic number is 12, therefore, there are 12 protons; the nuclide is charged (it is a cation with +2 charge, it has lost two electrons), the number of electrons is 10 (i.e 12 – 2 = 10); number of neutrons = 24 – 12 = 12

Isotopy is a phenomenon whereby atoms of the same elements have the same atomic number but different mass number. That is, isotopes are atomic species of the same element with the same number of protons but different number of neutrons. Isotopes of the same element have the same properties because they contain equal number of electrons. Examples of isotopes are: ${}^{1}_{1}H$, ${}^{2}_{1}H$, ${}^{3}_{1}H$; ${}^{16}_{8}O$, ${}^{17}_{8}O$, ${}^{18}_{8}O$ etc. Most elements exist as mixture of isotopes. The abundance of each mixture is called the isotopic abundance. The isotopic abundance can be expressed as percentage, which will add up to 100% and as fractions that will add up to 1. The relative abundance of isotopes in a mixture can be expressed as isotopic ratio. For example, a sample Br contains 50.69% ^{79}Br atoms and 49.31% ^{81}Br atoms; the isotopic ratio $\frac{^{79}Br}{^{81}Br}$ is $\frac{50.69}{49.31}$=1.028.

Elements, such as fluorine, sodium, aluminium, phosphorus and iodine that contain only one type of atom are isotopically pure.

Relative Atomic Mass

Atomic mass unit (amu) is defined as $\frac{1}{12th}$ of the mass of ^{12}C nuclide so that it has the value of 1.660540×10^{-27} kg. The masses (in amu) of the three fundamental particles have been presented in **Table 1.1**. The isotopic mass of any nuclide can be calculated using the masses of the fundamental particles. For example, ^{35}Cl is the sum of the masses of 17 protons, 18 neutrons and 17 electrons.

17(1.0007276 amu) + 18(1.0008665 amu) + 17(0.000549 amu) = 35.288995 amu (calculated value). The experimental value of the isotopic mass of ^{35}Cl is 34.96885 amu. The difference between the experimental and calculated values is 0.32015. This difference is regarded as *mass defect* or *mass deficit*.

The relative atomic mass (RAM) of an element is defined as the mass of one atom when compare to carbon–12 isotope. It the number of times one atom of the element is as heavy as one-twelfth of one atom of carbon–12. It is represented by A_r. Relative atomic mass has no unit. Therefore, on the RAM scale, isotopic masses are divided by one-twelfth of the mass of the ^{12}C nuclide. The relative isotopic mass of ^{35}Cl is $\frac{34.96885 \text{ amu}}{\frac{1}{12} \times 12.0000 \text{ amu}} = 34.96885$

The relative atomic mass is calculated by multiplying the relative isotopic mass of each isotope by its percentage abundance and adding all these values.

Example 1.2: calculate the relative atomic mass for naturally occurring chlorine if the distributions of isotopes are 75.77% ^{35}Cl and 24.23% ^{37}Cl. Accurate masses (relative isotopic masses) for ^{35}Cl and ^{37}Cl are 34.97 and 36.97 amu, respectively.

Solution:

$$A_r = \left(\frac{75.77}{100} \times 34.97\right) + \left(\frac{24.23}{100} \times 36.97\right) = 35.45$$

Example1.3: three isotopes of Mg occur in nature. Their abundances and masses are shown below:

Isotope	% Abundance	Mass (amu)
$^{24}_{12}Mg$	78.99	23.98504
$^{25}_{12}Mg$	10.00	24.98584
$^{26}_{12}Mg$	11.01	25.98259

Use the data in the table above to calculate the relative atomic mass of Mg.

Solution:

$$A_r = 0.7899(23.98504) + 0.10(24.98584) + 0.1101(25.98259) = 24.30$$

Example 1.4: the atomic weight of Ga is 69.72 amu. The masses of the naturally occurring isotopes are 68.9257 amu for $^{69}_{31}Ga$ and 70.9249 amu for $^{71}_{31}Ga$. Calculate the percentage abundance of each isotope.

Solution:

Let x represents the fraction $^{69}_{31}Ga$

Then $(1-x) =$ fraction of $^{71}_{31}Ga$

$\therefore\ x(68.9257) + (1-x)(70.9249) = 69.72$

$x = 0.6;$ then $1 - x = 1 - 0.6 = 0.4$

% of $^{69}_{31}Ga = 60\%$ while % of $^{71}_{31}Ga = 40\%$

Mass Spectrometry

Mass spectrometers (spectrophotometers) are instruments that measure the charge-to-mass ratio of charged particles. The technique of measuring the charge-to-mass ratio of charged particles using mass spectrometer is known as mass spectrometry. The technique is used to determine the relative atomic mass, masses of isotopes and isotopic abundance (relative amounts of the isotopes). It is a versatile technique for determination of relative molecular masses, identities and structures of organic compounds.

The basic features of mass spectrometers are shone in **Figure 1.4**. The sample if not in gaseous form, is first vaporised. In mass spectrometry, a gas sample at very low pressure is bombarded (in the ionisation chamber) with high-energy electrons, which causes electrons to be ejected from some of the gas molecules, creating positive ions. The positive ions are then focused into a very narrow beam, accelerated by an electric field and deflected by along a circular path by a magnetic field. The speed of the particles or ions depend on masses of the particles, charges on the particles, magnetic field strength and magnitude of the accelerating voltage. The lighter the particle, the greater the deflection. The intensity of ion beam is detected electrically, amplified and recorded.

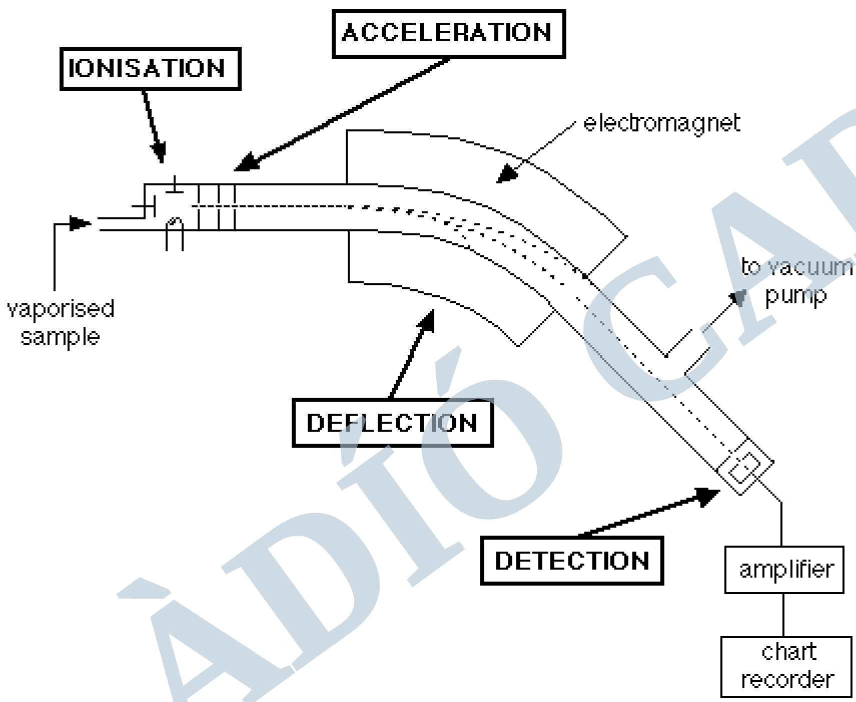


***Figure 1.4**: the basic features of a mass spectrometer*

The mass spectrum is produced by increasing the magnetic field so that ions of higher mass to charge ratios are gradually brought into line with the detector. The mass spectrum is usually calibrated using carbon–12 so that relative isotopic masses can be read directly. The relative heights give the isotopic abundances. Part of a mass spectrum of a mixture containing traces of chlorine atoms is shown in **Figure 1.5**. The spectrum shows that the abundances of ^{35}Cl and ^{37}Cl are 75 and 25%, respectively. The mass spectrograph operates on the same principle as a mass spectrometer. The main difference is that a mass spectrograph uses a photographic plate for detection instead of an electric device.

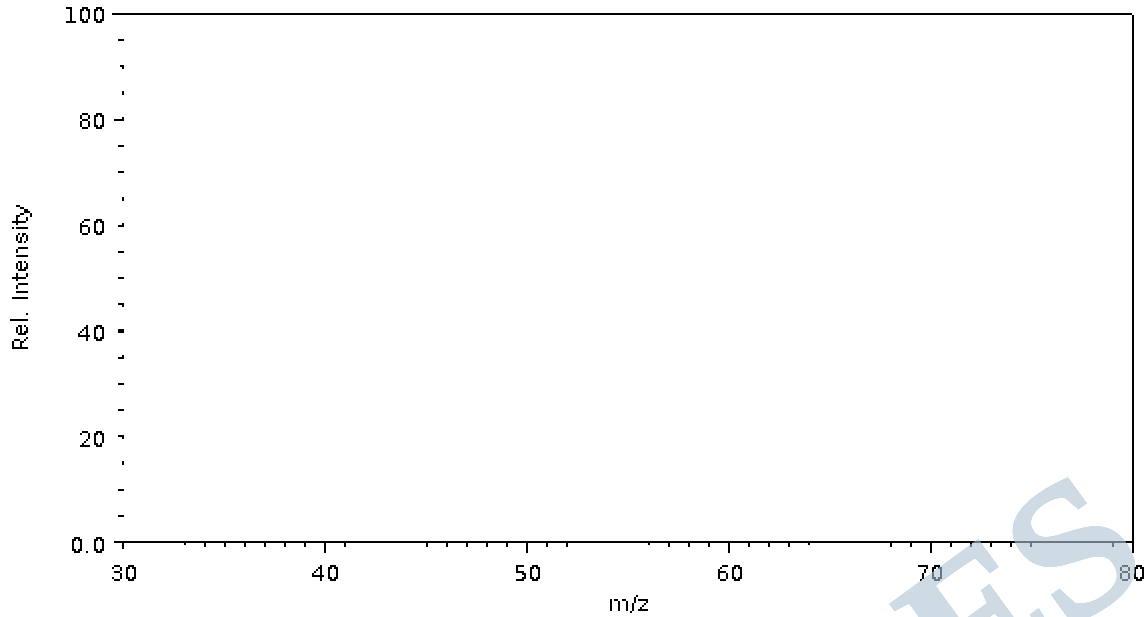


Figure 1.5*: the mass spectrum of chlorine*

Quanta and Photons

In 1900, Planck suggested that energy could only be absorbed or emitted in discrete quantities called *quanta*. A light ray is a stream of energy emitted by the source. Einstein had postulated that light consists of photons; light is regarded as being made up of discrete packets of electromagnetic energy. These packets are known as quanta or photons. The more intense the light, the greater the number of photons passing a given point. A dim source of light emits relatively few photons while a bright source of light emits a dense stream of photons. Experiments have shown that the energy (E) of a photon is proportional to the frequency (v) of radiation but inversely proportional to the wavelength (λ) of radiation, as shown in Equation 1.1.

$$E = h\upsilon = \frac{hc}{\lambda} \qquad 1.1$$

where h is the Planck's constant (6.626×10^{-34} J s); c is the speed of light (3×10^{8} m s^{-1}).

The wavelength is the distance between any two adjacent identical points of the wave, for instance, two adjacent crests or two adjacent troughs. The frequency is the number of wave crests passing through a given point per unit time; frequency is measured in s^{-1} or *Hz*. The wavelength and frequency are inversely proportional to each other.

The Electronic Structure of Atoms

Atomic Spectra

The fundamental measurement obtained in spectroscopy is a *spectrum*. A spectrum is a plot of measured light intensity versus some physical properties of light such as wavelength (λ), wavenumber ($\tilde{\upsilon}$) etc. It is a display or dispersion of the component of radiation. Spectrometer/spectrophotometer is used to measure a spectrum electronically. The components of full electromagnetic radiation are γ-rays, X-rays, ultraviolet, visible light, infrared, microwave and radio wave. **Figure 1.6** shows the electromagnetic spectrum.

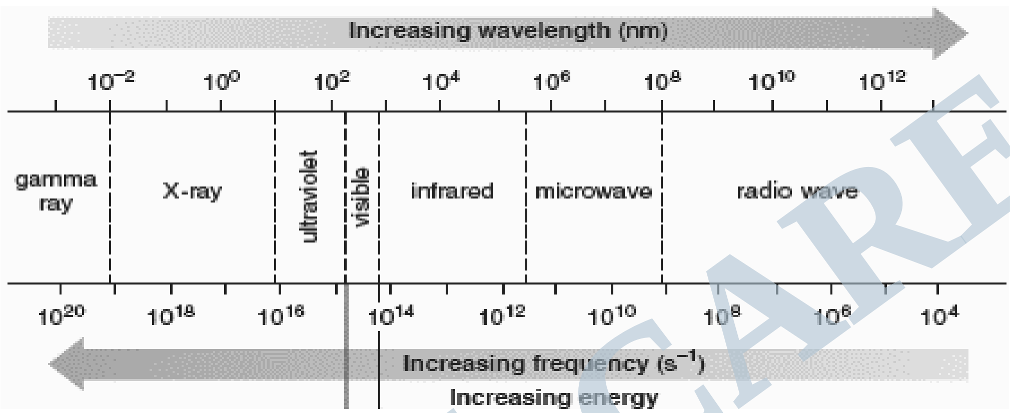


***Figure 1.6**: the electromagnetic spectrum*

Continuous radiation is a kind of radiation containing radiation of all wavelengths within a specific range. Its spectrum is called a continuous spectrum. The rainbow is a natural example of continuous spectrum. When a beam of continuous radiation such as white light is passed through a gaseous sample, the radiation emitted has certain wavelengths missing. The spectrum of this radiation is called an *atomic absorption spectrum*. The wavelengths of the radiation that has been absorbed by the atoms show up as dark lines on the continuous spectrum. When elements in their gaseous states are heated to high temperatures or subjected to electrical discharges, radiation of certain wavelengths is emitted. The spectrum of this type of radiation is known as *atomic emission spectrum*. An absorption or emission spectrum, which consists of lines, is called a discontinuous or line spectrum. Any part of the spectrum, where the lines converge is called a *continuum*.

Niels Bohr related the line in the atomic spectra to changes in energies of electrons in atoms. Electrons in atoms could only have certain fixed energy values. To each energy was given a number called *quantum number*. An electron could jump from one energy level to another by emitting or absorbing a fixed amount of energy. Emission of radiation occurs when electron moves from higher energy level to lower energy level while absorption takes place on moving from lower energy level to higher energy level. Each line in atomic spectrum corresponds to electrons moving from one specific energy level to another specific energy level. In other way, each spectral line corresponds to a specific wavelength of light or to a specific amount of energy that is either absorbed or emitted.

Fluorescent is a type of luminescence, which is the radiative transition from an excited state of the same spin multiplicity as the lower state. Fluorescence is allowed and occurs fast. If the electron spin multiplicitiesof the emitting and final states differ, the emission is known as phosphorescence. Phosphorescence is weakly allowed and the phosphorescence emission is less intense and less rapid than fluorescence.

Stimulated emission occurs when electrons in an excited state falls back to lower energy level and releases photo of characteristic wavelength. The word laser is an acronym for light amplification by stimulated emission of radiation. **Figure 1.7** represents a laser system

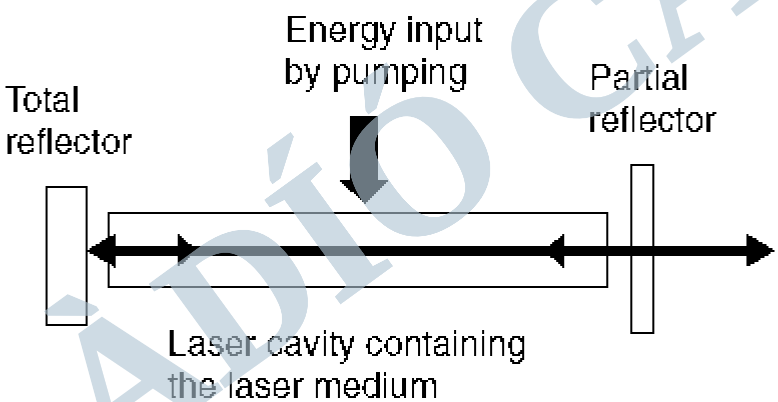


Figure 1.7*: laser system*

A flash of ultraviolet light from flash lamp excites electrons in the rod (**Figure 1.7**). A few of the excited electrons immediately and spontaneously fall back to lower energy levels and emit photons in the process. These photons are reflected back into the rod by the mirrors at either end. This stimulates a chain reaction whereby all the remaining excited electrons simultaneously fall back to their lower energy states. This produces a highly intense pulse of light with a specific direction and frequency. One of the mirrors is partially transparent thus allows the laser pulse to pass through.

Atomic Emission Spectrum of Hydrogen

Incandescent solids, liquids and high-pressure gases produce continuous spectra. When electric current is passed through hydrogen gas at very low pressure, several series of lines in the spectrum of hydrogen is produced. The atomic emission of hydrogen spectrum is shown in **Figure 1.8**.

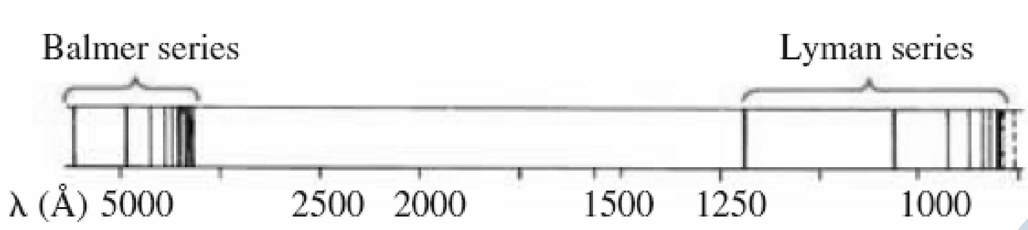


***Figure 1.8**: the atomic emission spectrum of hydrogen*

Many scientists studied these lines intensively. Johann Balmer and Johannes Rydberg proved that the wavelengths various lines in the hydrogen spectrum can be represented by a mathematical equation shown in Equation 1.2. The equation is an empirical equation.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad 1.2$$

where λ is the wavelength (m); R is a constant known as Rydberg constant ($1.097 \times 10^7\ m^{-1}$); n's are positive whole numbers; n_1 is smaller than n_2.

In 1913, Niel's Bohr, a Danish scientist, provided an explanation for Balmer and Rydberg observations. He wrote equation that described the electron of hydrogen atom as revolving round its nucleus in circular orbit. Electronic energy is quantised (only certain values of electronic energy are possible). He suggested that the electrons can only be in certain discrete orbit, and they absorb or emit energy in discrete amount as they move from one orbit to another. Each orbit corresponds to energy level. When an electron moves from a higher energy level to a lower energy level, it emits energy. However, when electron moves from a lower energy level to the higher energy level, it absorbs energy. The values of n_1 and n_2 in the Balmer-Rydberg equation signify the lower and higher energy levels, respectively. Bohr was able to determine two significant aspects of each allowed orbits:

(i) where the electron can be found with respect to the nucleus, that is, the radius (r) of the circular orbit.

$$r = n^2 a_o \qquad 1.3$$

where n is positive integers, and a_o is the Bohr radius ($a_o = 5.292 \times 10^{-11} m$).

(ii) how stable the electron would be in that orbit, that is, its potential energy (E).

$$E = \frac{1}{n^2}\left(\frac{h^2}{8\pi^2 m a_o^2}\right) = -\frac{2.180 \times 10^{-18}}{n^2} J \qquad 1.4$$

where h = Planck's constant ($h = 6.626 \times 10^{-34}$ J s); m = mass of the electron.

Lyman series (ultraviolet region) corresponds to electrons falling from higher energy levels to its ground state (quantum number 1; n_1 is 1). Balmer series corresponds to electrons falling from higher energy levels to the first excited state (quantum number 2; n_1 is 2). Paschen series (infrared region) corresponds to electrons falling from higher energy levels to the quantum number 3 (n_1 is 3). The two series in the far infrared region are Bracket and Pfund series that correspond to electron transition from higher energy levels to quantum numbers $n_1 = 4$ and $n_1 = 5$, respectively.

As the quantum numbers of the energy levels increase, the energy levels become closer and closer together until they converge. The convergence limits correspond to transition of electrons in these highest levels. The energy required to lift an electron from its ground state and detach it from its atom is known as ionisation energy. The frequency of radiation at the convergence limit can be converted to ionisation energy by using Equation 1.5.

$$\Delta E = hv \qquad 1.5$$

Lyman series corresponds to electron falling back to the ground state and the convergence limit of the series is used for calculating the ionisation energy. The first ionisation energy is defined as the minimum energy required to remove one mole of electrons from a gaseous atom of an element. The energy required to remove one mole of electrons from one mole of gaseous singly charged positive ions is known as the second ionisation energy.

Example 1.5: the convergence limit of the Lyman series has a wavelength of 91.4 nm, which corresponds to a frequency of 32.8 x 10^{14} Hz. Use this frequency to calculate the value of ionisation energy of one mole of hydrogen atoms (Planck's constant, h = 6.626 x 10^{-34} J s).

Solution:

The ionisation energy for one atom of hydrogen can be calculated using Equation 1.5.

$$\Delta E = hv = (6.626 \times 10^{-34}\, J\ s) \times (32.8 \times 10^{14}\, s^{-1}) = 2.17 \times 10^{-18}\, J$$

One mole of hydrogen contain 6.022 x 10^{23} atoms. Thus, the first ionisation energy (ΔH^o) for hydrogen is

$$\Delta H^o = (6.022 \times 10^{23}\, mol^{-1}) \times (2.17 \times 10^{-18}\, J) = 1306774 \text{ J mol}^{-1} = 1307 \text{ kJ mol}^{-1}$$

The Dual Nature of the Electron

The Bohr model is useful in explaining the lines in the hydrogen spectrum. It is also useful for explaining the arrangement of elements in the periodic table and the patterns in the ionisation energies of elements. However, Bohr model cannot be used to account for some of the more complex features of the spectra of the elements other than hydrogen. Secondly, there is no experimental evidence to show that an electron rotates with a fixed angular momentum in an orbit around the nucleus, and even if did, the electron would gradually lose energy and slow down, it would then be drawn in towards the nucleus and this would lead to the collapse of the atom.

Einstein suggested that light can exhibit both wave and particle properties while Louis de Broglie (1892–1987) also suggested that very small particles, such as electrons, might also display wave properties. In de Broglie doctoral thesis in 1925, he predicted that a particle with a mass (m) and velocity (v) should have a wavelength (λ) associated with it. The mathematical representation of the de Broglie wavelength (Equation 1) is given below:

$$E = h\upsilon = \frac{hc}{\lambda}$$

Einstein's equation is presented in Equation 1.6.

$$E = mc^2 \qquad 1.6$$

Using Equations 1.1 and 1.6, Equation 1.7 (de Broglie equation) is obtained.

$$\lambda = \frac{h}{mv} \qquad 1.7$$

Example 1.6: (a) a green light of wavelength 4.8 x 10^{-7} m is observed in the emission spectrum of hydrogen. Calculate the energy of one photon of this light.

(b) calculate the wavelength of an electron travelling at 72 km h^{-1}. Note: Planck's constant, h = 6.626 x 10^{-34} J s; mass of electron, m_e = 9.11 x 10^{-31} kg; velocity of light, c = 3.0 x 10^8 m s^{-1}.

Solution:

(a) Using $E = \frac{hc}{\lambda}$

$$E = \frac{(6.626\times10^{-34}\, J\ s)\times(3.0\times10^{8}\, m\ s^{-1})}{4.8\times10^{-7}\, m} = 4.14\times10^{-19}\, J$$

(b) Using $\lambda = \frac{h}{mv}$

The velocity, $v = 72\ km/h = \frac{72\times1000\ m}{3600\ s} = 20\ m\ s^{-1}$

$$\lambda = \frac{6.626\times10^{-34} kg\ m^2\ s^{-1}}{9.11\times10^{-31} kg \times 20\ m\ s^{-1}} == 3.64\times10^{-5} m$$

Orbitals

It is not possible to determine accurately both the momentum and the position of an electron (or any other very small particle) simultaneously. This is the Heisenberg uncertainty principle. Momentum is mass times velocity, *mv*. However, although the position of an electron cannot be determined exactly, the probability of finding an electron in a certain position at any time can be ascertained with the aids of statistical approach. A region or volume of space within which there is a high probability of finding an electron is known as an orbital. Orbital corresponds to a stable energy state for the electron and is described by a particular set of quantum numbers. Electron can occupy four types of orbitals: *s*, *p*, *d* and *f* orbitals. The *s* orbital is spherical while *p* orbital has dump-bell shape as shown in **Figure 1.9**. Electron has a negative charge, thus its orbital can be regarded as a spread of charge. This is often called a charged cloud.

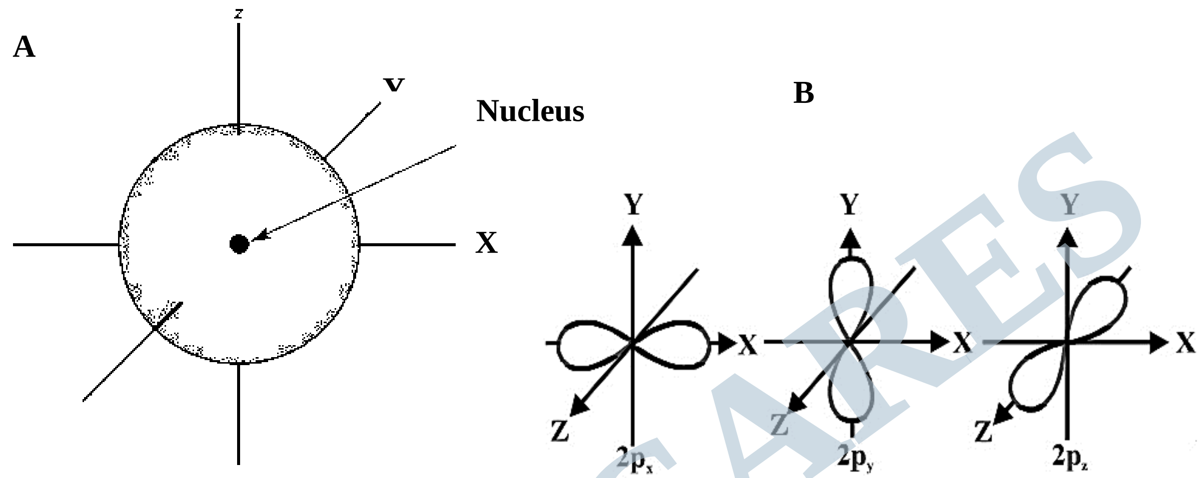


***Figure 1.9**: the shapes of **A**) s-orbital, and **B**) p-orbital*

Quantum Numbers

Quantum numbers are used to represent the electronic arrangement of atoms (electronic configuration). The solutions of the Schrödinger and Dirac equations for hydrogen atoms gave wave functions, ψ, that describe the various states available to hydrogen's single electron. Each of these possible states is described by four quantum numbers. The quantum numbers help to know the energy levels of electrons and the shapes of the orbitals that describe distributions of electrons in space.

(i) The principal quantum number, *n*, specifies the energy level or shell, which an electron occupies. This quantum number may be any positive integer; $n = 1, 2, 3, 4$……

The lowest energy level is obtained when $n = 1$. This lowest energy state is called the ground state of the atom. The orbitals form a series of thick shells, resembling fuzzy layers of an imaginary onion. Shells of higher *n* surround the inner shells of lower *n*. The shells are named *K*, *L*, *M*, and *N* to represent the quantum numbers $n = 1, 2, 3$, and 4, respectively. The numbers represent increasing energy levels of the shells. For a given *n*, the electron capacity in a shell is calculated using $2n^2$.

(ii) The Azimuthal (orbital or subsidiary or angular momentum) quantum number, l, governs the shape of the orbital. The angular quantum number designates a sublevel or a specific shape of atomic orbital that an electron can occupy. This quantum number can have integral values ranging from 0 up to (n – 1).

l = 0, 1, 2, 3, ……, (n – 1)

The maximum value of l is (n – 1). A letter notation, which corresponds to a different sublevel (subshell) and orbital shape is given to l.

$$\begin{array}{l} l = 0,\ 1,\ 2,\ 3,\ 4,\ 5......(n-1) \\ \quad\ \ s\ \ p\ d\ \ f\ \ g\ \ h \end{array}$$

All the orbitals with a given value of the Azimuthal quantum number, l, are said to belong to the same subshell of a given shell. In the first shell (n = 1), the maximum value of l is zero, which signifies that there is only an s subshell and no p subshell. However, in the second shell (n = 2), the permissible values of l are 0 and 1, which means that there are s and p subshells. All orbitals in the same subshell degenerate, that is, they are of the same energy.

(iii) The magnetic quantum number, m_l, labels the different orbitals within a subshell. Orbitals within a given subshell have different orientations in space, but their energies are the same. Within each subshell, m, takes any integral values from $-l$, through zero up to and including $+l$. The maximum value of m_l depends on the value of l.

m_l = ($-l$), ……. 0, ……… ($+l$)

For example, for the subshell with l = 1, which consists of p-orbital, m_l can have the values of -1, 0, +1; so there are three p-orbitals (p_x, p_y, and p_z orbitals).

(iv) The spin quantum number, m_s, refers to the spin of an electron and the orientation of the magnetic field produced. Each atomic orbital can accommodate only two electrons. An electron has two spin states represented by two arrows $(\uparrow\downarrow)$, one clockwise at a certain rate and the other is counter-clockwise at exactly the same rate. In summary, for every set of n, l, and m_l values, m_s can take the value of $+\frac{1}{2}$ or $-\frac{1}{2}$.

Therefore, $m_s = \pm\frac{1}{2}$

Table 1.2 shows the permissible values of quantum numbers 1 to 4. The sizes of orbitals increase as n increases. The following statements are valid for orbitals:

(i) all orbitals with the same principal quantum number n are similar in size in any atom;

(ii) the larger values of n correspond to larger orbital size in any atom; and

(iii) each orbital with a given n value becomes smaller as nuclear charge increases.

***Table 1.2**: the permissible values of quantum numbers 1 – 4.*

n	l	m_l	m_s	Electron capacity of subshell ($4l+2$)	Electron capacity of shell ($2n^2$)
1	0 (1*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	2
2	0 (2*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	8
	1 (2*p*)	–1, 0, +1	$\pm\frac{1}{2}$ for each value of m_l	6	
3	0 (3*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	18
	1 (3*p*)	–1, 0, +1	$\pm\frac{1}{2}$ for each value of m_l	6	
	2 (3*d*)	–2, –1, 0, +1, +2	$\pm\frac{1}{2}$ for each value of m_l	10	
4	0 (4*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	32
	1 (4*p*)	–1, 0, +1	$\pm\frac{1}{2}$ for each value of m_l	6	
	2 (4*d*)	–2, –1, 0, +1, +2	$\pm\frac{1}{2}$ for each value of m_l	10	
	3 (4*f*)	–3, –2, –1, 0, +1, +2, +3	$\pm\frac{1}{2}$ for each value of m_l	14	

Electronic configuration

This is defined as the arrangement of electrons into their shells, sub-shells and orbitals. The term normally applies to atoms in their ground states. The ground state electronic configuration corresponds to the electronic configuration of an isolated atom in its ground or unexcited state. The electronic configuration of an atom with one or more electrons in an excited state is termed an excited-state electronic configuration. There are three rules for determining the exact ground state electronic configuration of elements; the Aufbau Principle, the Pauli Exclusion Principle, and the Hund's Rule.

Aufbau Principle

Aufbau is a German word for 'building up'. Aufbau principle states that the electrons in their ground states occupy orbitals in order of the orbital energy levels. The lowest energy orbitals are always filled first. When describing a ground state electron configuration, it must be put in mind that the total energy of the atom must be kept as low as possible. The orbitals increase in energy as the value of n increases. For a given value of n, energy increases with increasing value of l. Therefore, within a particular shell, the *s* subshell has the lowest in energy, then *p* subshell, then the *d* subshell, then the *f* subshell, and so on. However, the order of energies of the orbitals somehow vary from atom to atom as result of changes in the nuclear charge and interactions among the electrons in the atom.

There are two generalised rules for predicting electronic configurations. The two rules are:

(i) electrons are assigned to orbitals in order of increasing value of $(n+l)$; and

(ii) electrons are assigned first to the subshell with lower n for subshells with the same value of $(n+l)$.

Illustrations for rule (i): the 2*s* subshell has $(n + l = 2 + 0 = 2)$ and the 2*p* subshell has $(n + l = 2 + 1 = 3)$, it is expected therefore that the 2*s* subshell will be filled before filling 2*p*. Similarly, 4*s* subshell $(n + l = 4 + 0 = 4)$ will be filled before filling 3*d* subshell $(n + l = 3 + 5 = 5)$.

An illustration for rule (ii): 2*p* subshell $(n + l = 2 + 1 = 3)$ is filled before 3*s* $(n + l = 3 + 0 = 3)$ because 2*p* has a lower value of n.

Pauli Exclusion Principle

This principle state that no two electrons in an atom may have identical sets of four quantum numbers. Pauli Exclusion Principle governs the electronic structures of atoms. It means that each orbital can hold no more than two electrons and that the two electrons in an orbital must have opposite spins. As shown in **Table 1.2**, the maximum number of electrons that can be accommodated in the subshells *s*, *p*, *d* and *f* are

2, 6, 10 and 14, respectively. The maximum number of electrons that *K*, *L*, *N* and *M* shells can accommodate are 2, 8, 18 and 32, respectively. These values can be gotten by using $2n^2$, where n is the principal quantum number.

Hund's Rule

Hund's rule states that electrons occupy all the orbitals of a given subshell singly and with parallel spins before pairing begins. Carbon has two unpaired electrons in its 2*p* orbitals whereas nitrogen has three.

A simple diagram (**Figure 1.10**) can be used to illustrate the order in which the subshells are occupied with electrons.

1*s*
2*s* 2*p*
3*s* 3*p* 3*d*
4*s* 4*p* 4*d* 4*f*
5*s* 5*p* 5*d* 5*f* 5*g*
6*s* 6*p* 6*d* 6*f* 6*g* 6*h*
7*s* 7*p* 7*d* 7*f* 7*g* 7*h*

***Figure 1.10**: diagram for filling subshells with electrons*

The order in which atomic orbitals are occupied according to the Aufbau principle is given in **Table 1.3**. Each time an electron is added, we move one place to the right. At the end of a period, we move to the beginning of the next period.

For transition metals, the electrons are added to the *d*-orbitals as the atomic number increases. The electronic configurations of Cr (Z = 24; $[Ar]3d^5 4s^1$) and Cu (Z = 29; $[Ar]3d^{10} 4s^1$) are exceptionally written because half-filled and fully-filled orbitals are stable more than any other configuration. The half-filled subshell configuration (d^5) and the completely filled subshell (d^{10}) experimentally have low energy values. The transition metals in the same period differ mainly in the number of *d*-electrons; therefore, their properties are similar. Most of the transition metals form ions in more than one oxidation state because the *d*-electrons have similar energies and variable number can be lost. Electrons occupy 4*p*-orbital once the 3*d*-orbitals are filled-up.

For Period 5 elements, the 5s-orbital is first occupied before the 4d-orbital. In Period 4, the energy of the 4d-orbital falls below that of the 5s-orbital after two electrons have been accommodated in the 5s-orbital. A similar phenomenon is observed in Period 6, but another set of inner orbitals, the 4f-orbitals, begins to be occupied. Cerium, for example, has configuration $[Xe]4f^1 5d^1 6s^2$. Electrons begin to occupy the seven 4f-orbitals, which are complete after 14 electrons have been added. For ytterbium, we have $[Xe]4f^{14}6s^2$. Next the 5d-orbital is occupied. The 6p-orbital is occupied only after the 6s- and 5d-orbitals are filled for mercury. The electronic configuration of thallium is $[Xe]4f^{14}5d^{10}6s^2 6p^1$.

***Table 1.3**: building up of atomic orbitals*

Atomic Number (Z)	**Element**	**Symbol**	**Configuration**	**Simplified configuration**
1	Hydrogen	H	$1s^1$	
2	Helium	He	$1s^2$	
8	Oxygen	O	$1s^2 2s^2 2p^4$	$[He]2s^2 2p^4$
18	Argon	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$	
19	Potassium	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	$[Ar]4s^1$
20	Calcium	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	$[Ar]4s^2$
21	Scandium	Sc	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$	$[Ar]3d^1 4s^2$
24	Chromium	Cr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$	$[Ar]3d^5 4s^1$
25	Manganese	Mn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$	$[Ar]3d^5 4s^2$
29	Copper	Cu	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$	$[Ar]3d^{10} 4s^1$
30	Zinc	Zn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$	$[Ar]3d^{10} 4s^2$

Example 1.7: (i) Write the ground state electronic configuration of nitrogen.

(ii) How many unpaired electron(s) has the oxygen atom?

(iii) Write the ground state electronic configuration of an element with Z = 33.

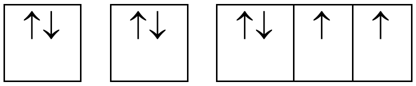
(iv) Write the electronic configuration of Al^+ .

Solutions:

(i) The atomic number of nitrogen is 7, the electronic configuration is written as:

$[He]2s^2 2p^3$ or $1s^2 2s^2 2p^3$ or $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

(ii) The electronic configuration of oxygen is $1s^2 2s^2 2p^4$.



There are two unpaired electrons in the oxygen atom.

(iii) The ground state electronic configuration of an element with an atomic number 33 is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^3$

(iv) The atomic number of Al is 13.

Al^+ has lost one electron from the outermost shell. Therefore, the electronic configuration of Al^+ is $1s^2 2s^2 2p^6 3s^2$

The Periodicity of Atomic Properties

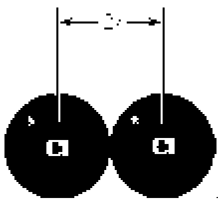
The properties that arise from the elucidation of atomic structure depends on the structure of the atom. Examples of these properties are electronic configurations, atomic radius, atomic number, ionic radius, ionisation energy, electron affinity, electronegativity, etc. A good knowledge of periodicity is important for understanding the bonding in simple compounds. Many physical properties, such as melting point, boiling points, and atomic volumes, show periodic variations. The variations largely depend on the electronic configuration in the outermost occupied shell and how far that shell is away from the nucleus.

Atomic Number and Atomic Radius

Atomic number represents the number of protons in the nucleus of an atom or it is the unit positive charge of the proton or it is the number of external electrons in a neutral atom. Atomic number is the order in which elements appear in the periodic table. Atomic number increases down the group and across the period.

Atomic radius is the distance from the nucleus to the boundary of the surrounding cloud of electrons. It can also be defined as the half the distance between the centres of the neighbouring atoms. Atomic radius is a measure of the size of the atom of an element. It is measured in Angstrom (Å). If the element is a metal, then the distance is between the centres of the neighbouring atoms in a solid sample. If the element is a non-metal, then the appropriate distance is the distance between the centres of atoms joined by a chemical bond; this radius is also called the covalent radius of the element. The distance between the nuclei in Cl_2 molecules is 198 pm (picometre; 10^{-12} m), so the atomic (covalent) radius of chlorine atom is 99 pm.

Within a family of representative elements, atomic radii increase down the group as electrons are added to the shells farther from the nucleus. Moving from left to right across the period of the periodic table, the atomic radii of representative elements decrease as a proton is added to the nucleus and an electron is added to a particular nucleus.


Atomic radius

The radius of an atom depends on two factors:

(i) the attraction of the outer electrons by the inner electrons.

(ii) the screening or shielding effect of outer electrons by the inner electrons. Effective nuclear charge (Z_{eff}) is experienced by an electron in an outer shell is less than the actual nuclear charge, Z. This phenomenon is as a result of the attraction of outer-shell electrons by the nucleus, which is partly counter-balanced by the repulsion of the outer-shell electrons by electrons in inner shells. The electrons present in the inner shells screen or shield the electrons in outer shells from the complete effect of the nuclear charge. For instance, Li has two electrons in a filled *s*-shell, $1s^2$, and one electron in the 2*s* orbital, $2s^1$. The one electron in the 2s orbital is fairly effectively screened from the nucleus by the two electrons in the filled 1*s* orbital. This therefore does not make the 2*s* electron "feel" the full 3+ charge of the nucleus. Sodium (Z = 11) has ten electrons in inner shells, $1s^2 2s^2 2p^6$, and one electron in an outer shell, $3s^1$. The ten inner-shell electrons of the sodium atom shield the outer-shell electron from most of the 11+ nuclear charge. The third shell (n = 3) is farther from the nucleus than the second shell (n = 2). This is why sodium atoms are larger than lithium atoms.

Across the periodic table (from left to right), atoms become smaller because the increasing effective nuclear charges increase even though more electrons are being added to the shells. For the transition elements, the variations are irregular because electrons are added to an inner shell, however, the general trend of decreasing radii certainly continues as we move across the *d*- or *f*-block metals. All transition elements have smaller radii than the preceding Group 1A and 2A elements in the same period.

Ionic Radii

The ionic radius is defined as equilibrium distance between two similar atoms joined together by a single covalent bond. Ionic radius of an element is its share of the distance between neighbouring ions in an ionic solid. The distance between the nuclei of a neighbouring cation and anion is the sum of the two ionic radii, in reality, we use the radius of the oxide ion, 140 pm, to calculate the radii of other ions. For example, the distance between Mg and O nuclei in MgO is 205 pm, the radius of Mg^{2+} ion is calculated as 205 – 140 = 65 ppm. Generally, ionic radii increase down a group but decrease across the period from left to right. At times, we compare atoms to their ions, atoms or ions that are vertically positioned on the periodic table, or isoelectronic species. It is correct to say that:

(i) cations are always smaller than the neutral atoms from which they are formed;

(ii) anions are always larger than the neutral atoms from which they are formed;

(iii) both cation and anion sizes increase going down a group;

(iv) within an isoelectronic series, ionic radii decrease with increasing atomic number because of increasing nuclear charge.

Metallic radius is the half of the internuclear orbital distance between two atoms of nearest neighbour. A metallic radius is always greater than the atomic radius or ionic radius by at least $\frac{1}{8}th$.

Ionisation Energy

Ionisation energy (IE) of an element is a measure of the relative tendency of an atom to attract electrons to itself when it is chemically combined with another atom. It is a measure of how tightly electrons are bound to atoms. The first ionisation energy (first ionisation potential) is defined as the minimum amount of energy required to remove the most loosely bound electron from an isolated gaseous atom to form an anion with a +1 charge.

$$Atom + energy \rightarrow cation + e^-$$

The second ionisation energy is the amount of energy required to remove the second electron. Second ionisation energy is always greater than the first ionisation energy because it is always difficult to remove an electron from a positively charged ion than from the corresponding neutral atom. Ionisation always requires energy to remove an electron from the attractive force of the nucleus.

Low ionisation energy means that it is easy to remove electron, hence ease of positive ion (cation) formation. Elements with low ionization energies will easily lose electrons to form cations. Ionisation energy could also be described as a measure of metallic character of element. The ionisation energies of metals are below 800 kJ/mol while the ionisation energies of non-metals are above the value. Metals have low ionisation energies and could easily lose their electrons.

Ionisation energies usually decrease down the group. This because the outermost electron occupies a shell that is farther from the nucleus and is therefore less tightly held. With few exception, first ionisation energy increases from left to right across the period. The effective nuclear charge increases as we move from left to right across a given period. As a result, the outermost electron is gripped more tightly and ionisation energies usually increase. Ionisation energy depends on three factors:

(i) nuclear charge;

(ii) the distance between the nucleus; and

(iii) the electron to be removed.

As said earlier, the nuclear charge increases across the period, then the radius decreases. These factors cause electrons to be held strongly to the nucleus and become more difficult to remove an electron, thus an increase in ionisation energy. Increase in ionisation energy is not uniform. In period 2, the trend is reverse in passing from Be to B and also from N to O. The major reason for this reversal is stability associated with *s*-subshell of Be and a single electron present in B. The stability is because of half-filled *p*-subshell. In the period 4 onwards, the general trend is interrupted by the transition metals.

Electron Affinity

Electron affinity (EA) is defined as the energy that is released when a neutral atom gained an electron or it is the energy change that occurs when one mole of singly charged anion is formed from one mole of gaseous atom.

$$Cl_{(g)} + e^- \rightarrow Cl^-_{(g)} + 349\ kJ \qquad \text{EA} = -349\ \text{kJ/mol}$$

Electron affinity is the addition of an electron to a neutral gaseous atom or the process by which a neutral atom gains an electron. We normally assign a negative value for a release of energy and a positive value for absorption of energy. Some elements, such as noble gases, have no affinity for an additional electron and therefore have an electron affinity of zero. Most first electron affinities are negative; the more the negative value, the greater the electron affinity and thus the greater the ease of ions formation. The second electron affinity of an element is always endothermic because energy must be absorbed to overcome the repulsive forces between the two negative particles.

For example, $O^{-}_{(g)} + e^{-} \rightarrow O^{2-}_{(g)}$ 2nd EA= +791 kJ/mol

Electronegativities

Electronegativity (EN) of an element is defined as a measure of the tendency of an atom to attract electrons to itself when it is chemically combined with another atom. This term is the opposite of electropositivity. Non-metals have high electronegativities that is why they often gain electrons to form anions. The higher the electronegativity, the more stable the anions that are formed. Metals have low electronegativities and they often lose electrons to form cations. The lower the electronegativity, the more stable the cations that are formed. Electronegativities usually increase from left to right across periods but decrease down the groups. Variations among the transition metals are irregular, but they follow the same trends as the other representative elements.

Metals can lose electron(s) easily from their outermost shells, then they are said to be weakly electronegative (highly electropositive) while non-metals can gain electron(s) and are said to be strongly electronegative. Electronegativities of the elements are expressed on an arbitrary scale, called the Pauling scale. The simplest formula for estimating electronegativity is given by Equation 1.8.

$$\text{EN} = \frac{\text{effective nuclear charge}}{\text{covalent atomic radius}} \qquad 1.8$$

The electronegativity of fluorine (4.0) is higher than that of any other element. This means that fluorine has a greater tendency to attract electron density to itself than does any other element when fluorine is chemically bonded to other elements. There are exceptions to the general trend of electronegativities e.g. Group IIIA: Al (1.61); Ga (1.81); In (1.78) and Group IVA: Si (1.90); Ge (2.01); Sn (1.96).

Diagonal Relationship

The atomic radius and some chemical properties of some period 2 elements is similar to that of the element at their lower right in the periodic table. Vertical and horizontal trends describe the trend in chemical properties of element within the periodic table. The element at the head of each group also commonly possesses a diagonal relationship with the element to its lower right. Diagonal relationships arise because atomic radii, charge densities, electronegativities, and many chemical properties of the two elements are similar. The most noticeable diagonal relationship exists between Li and Mg. There is also strong diagonal relationship between Be and Al. So also between B and Si.

RADIOACTIVITY

Radioactivity is the spontaneous emission of radiation by an element due to the splitting of atomic nuclei. Examples of radioactivity elements are uranium, radium, polonium and actinium. The components of radiation from radioactive materials are alpha (α) particle, beta (β) particle and gamma (γ) ray. During radioactivity, one or more of these components are involved. Henri Becquerel discovered "radioactive rays" (natural radioactivity) emanating from a uranium compound. He was regarded as the father of radioactivity. Some years after Becquerel discovery, it was shown that nuclear reactions could be initiated by bombardment of nuclei with accelerated subatomic particles or other nuclei (induced or artificial radioactivity). Marie Curie and her husband showed that each of the components causes ionisation of gases and other substances. The ionising power of α-particle is greater than that of β-particle while the ionisation power of β-particle is greater than that of γ-ray.

The chemical properties are determined by electron distributions and are not directly influenced by atomic nuclei. Nuclear reactions are reactions that involve changes in the composition of nuclei. The major differences between ordinary chemical reactions and nuclear reactions are highlighted in **Table 1.4**.

***Table 1.4**: differences between ordinary chemical and nuclear reaction*

Ordinary Chemical Reaction	**Nuclear Reaction**
This reaction produces no new elements	There may be conversion of one element into another element
Only the electrons participate in the reaction	Particles within the nucleus participate in the reaction
Rate of chemical reaction are affected by external factors such as concentration, pressure, and temperature, catalyst	Rate of nuclear reaction is not affected by external factors
Small amounts of energy are released or absorbed	Enormous amounts of energy are released or absorbed

Alpha Particle

This consists of streams of high-velocity He nuclei. The nuclei has a mass of 4 and a charge of 2, and can be written as ${}^{4}_{2}He$ or He^{2+}. When the nuclide of radium-226 loses α-particle, the nuclide of radon-222 is formed. This could be written as:

$${}^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

The size of He nuclei limits the range and penetrating power of α-radiation compared with β-particle and γ-ray. The penetrating power of γ-ray is greater than that of β-particle while the penetrating power of β-particle is greater than that of α-particle.

Beta Particle

There are two types of β-radiation; β^- and β^+ radiations. The β^- radiation consists of streams of electrons moving at speeds comparable to the speed of light. An electron $\left({}_{-1}^{0}e\right)$ is emitted from an unstable nucleus when the ratio of neutrons to protons disproportionates. A neutron $\left({}_{0}^{1}n\right)$ splits up to produce an electron and a proton $\left({}_{1}^{1}p\right)$.

$${}_{0}^{1}n \longrightarrow {}_{1}^{1}p + {}_{-1}^{0}e$$

An additional proton is formed in the nucleus, then the atomic number is increased by one. A new element is thus formed. For example, when thorium-234 loses an electron, the nuclide of protactinium is formed.

$${}_{90}^{234}Th \longrightarrow {}_{91}^{234}Pa + {}_{-1}^{0}e$$

The β^+ radiation arises from emission of positron from a nucleus. A positron is similar to an electron except that it has a positive charge. It is formed by the conversion of a proton to a neutron:

$${}_{1}^{1}P \longrightarrow {}_{0}^{1}n + {}_{1}^{0}e$$

For example: ${}_{19}^{38}K \longrightarrow {}_{18}^{38}Ar + {}_{1}^{0}e$

Gamma Radiation

Gamma radiation is a high-energy electromagnetic radiation. Gamma rays are comparable to *X*–rays, although they have shorter wavelengths. Gamma rays have a long range and high penetrating power. They are emitted when a nuclide emits *α*– or *β*– particle. The *γ*–rays are not deflected by electric or magnetic field—meaning that *γ* rays have no charge. The isotope ${}^{119}Sn$ can be produced in metastable excited state, from which it decays with the emission of *γ*–rays.

$${}_{50}^{119}Sn \xrightarrow{\text{internal transition}} {}_{50}^{119}Sn + \gamma$$

***K*–capture Transformation**

Apart from the transformations resulting from the emission of alpha, beta and gamma radiations, another radioactive process is the capture by the nucleus of an electron from outside the nucleus. This process is termed *K*–capture and results in a decrease in atomic number but no change in mass number. An electron falling from a higher energy level is to replace the electron captured from the *K*–shell and this results in the emission of *X*–rays. For example:

$$^{133}_{56}Ba \xrightarrow{K-capture} {}^{133}_{56}Cs + X - ray$$

Detection and Measurement of Radioactivity

The activity of a radioactive material is the radioactivity per unit mass of the material. The level of activity is measured either in counts per minute (cpm) or counts per second (cps). Some of the methods used for detection and measurement of radioactivities are gas ionisation, scintillation methods, autoradiography and cloud chamber.

Gas Ionisation

The Geiger-Muller tube and counter rely on gas ionisation. The tube consists of an ionisation chamber filled with argon. The radiation ionises the argon to form the positive argon ions, which move to a cathode and the electrons to an anode. This creates an electrical pulse, which is amplified. The pulse registers as clicks, light flashes on meter reading.

Scintillation Methods

Some materials (fluorescent substances) absorb the high-energy of radiations such as gamma rays and convert it into light. As these materials absorb the radiation, the absorbing atoms jump to excited electronic states. The excited electrons return to their ground states through a number of transitions, some of which emit visible light. Zinc sulphide is an example of such materials. In a scintillation counter, the flashes of light cause pulses of electric current to pass through the photoelectric tube. They are amplified and counted.

Autoradiography

The radioactive sample is placed on a photographic emulsion containing silver halide. The radioactivity of the sample is estimated by developing the films and viewing it under high magnification. After a photographic plate is developed and fixed, the intensity of the exposed area is related to the amount of radiation that struck the plate. However, quantitative detection and analysis of radiation by this method is difficult and tedious.

Cloud Chambers

The cloud chambers was discovered by C.T.R. Wilson in 1911. The chamber contains air saturated with vapour. Particles emitted from a radioactive substance ionise air molecules in the chamber. When the chamber is cooled, droplets of liquid condense on the ions. The paths of the particles can be monitored by observing the fog-like tracks produced. The tracks can be photographed and analysed.

Stable and Unstable Isotopes

Nuclei whose neutron-to-proton ratios lie outside the stable region undergo spontaneous radioactive decay by emitting one or more particles or electromagnetic rays or both. Stable isotopes do not undergo radioactive decay. The majority of naturally occurring elements consist of one stable isotope in high abundance, along with small amount of other stable and unstable isotopes. Unstable isotopes are radioactive. Radioisotopes can be classified as natural or artificial. Both types of radioisotopes decay by emitting α– or β– particles. The decay continues until a stable isotope is formed. All naturally occurring isotopes with atomic number greater than 83 (bismuth) are unstable.

Half-life

The half-life ($t_{\frac{1}{2}}$) of an unstable isotope is the time taken for half the initial number of nuclei to disintegrate. Radioactive disintegration is of the first-order kinetics. The rate of decay is directly proportional to the number (N_t) of radioactive atoms present:

$$\frac{dN}{dt} \alpha N_t \qquad \text{therefore, } \frac{dN}{N_t} = kdt$$

$$\ln N_t = -kt + C, \quad \text{when t=0, N}_\text{t}\text{=N}$$

$$C = \ln N \quad then \quad \ln N_t = -kt + \ln N$$

$$\ln\left(\frac{N}{N_t}\right) = kt$$

At half-life, $N_t = \dfrac{N}{2}$

Therefore, $\ln\left(\dfrac{N}{N/2}\right) = kt_{\frac{1}{2}}$

Then $t_{\frac{1}{2}} = \dfrac{\ln 2}{k} = \dfrac{0.693}{k}$

Half-life does not depend on the initial number of nuclides. Both $t_{\frac{1}{2}}$ and k are characteristics of a particular isotope.

Decay Series

Thorium-232 decays by emitting an alpha particle: ${}^{232}_{90}Th \longrightarrow {}^{228}_{88}Ra + {}^{4}_{2}He$.

${}^{232}_{90}Th$ is termed the parent and ${}^{228}_{88}Ra$ is called the daughter. Radium-228 itself is unstable and decays by emitting a beta particle to form actinium-288.

$${}^{228}_{88}Ra \longrightarrow {}^{228}_{89}Ac + {}^{0}_{-1}e$$

The daughter then becomes a parent. The parent-daughter process continues until a stable isotope, which is lead-208 (${}^{208}_{82}Pb$) is formed. The complete process is known as a decay or disintegration series.

Artificial Nuclear Transformation

Both stable and unstable isotopes can be produced by bombarding nuclei with high-energy particles. For examples:

$${}^{14}_{7}N + {}^{4}_{2}He \longrightarrow {}^{18}_{8}O + {}^{1}_{1}H$$

$${}^{23}_{11}Na + {}^{1}_{1}H \longrightarrow {}^{23}_{12}Mg + {}^{1}_{0}n$$

Elements with atomic number 93 or more do not occur naturally. They have to be produced artificially by nuclear reactions and they are termed *trans-uranium* elements. For examples:

$$^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{239}_{92}U$$

$$^{239}_{92}U \longrightarrow ^{239}_{94}Pu + 2\,^{0}_{-1}e$$

$$^{238}_{92}U + ^{14}_{7}N \longrightarrow ^{252}_{99}Es$$

Most of the elements with atomic number 99 above were made by using bombarding projectile such as $^{14}_{7}N$. Elements with atomic number 101 upwards are known as the *trans-fermium* elements.

Nuclear Fission and Fusion

Binding energy is related to mass defects using Einstein equation ($E = mc^2$), where E is the binding energy, m is the mass, and c is velocity of light. Binding energy is usually expressed in MeV (million electron volts); 1 MeV = 9.6 x 10^{10} J/mol.

Nuclear fission is the breaking down of heavy nuclei into fragments. Isotopes of some elements with $Z >$ 80 are susceptible to fission in which they split into nuclei of intermediate masses and emit one or more neutrons. Some of these fission processes are spontaneous while others require that the activation energy be supplied by bombardment. For instance, fission can be induced by bombarding heavy nuclei with neutrons. Since two or more neutrons are emitted from every uranium-235 nucleus, a nuclear chain reaction is initiated and this results in an explosive chain reaction with the release of massive amount of energy. The complete chain reaction and loss of energy take place in less than one microsecond (10^{-6}). The energy that is produced or lost can be safely used as a heat source in a power plant.

For example, a possible fission of U-235 isotope is:

$$^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{92}_{36}Kr + ^{141}_{56}Ba + 3\,^{1}_{0}n$$

Nuclear fusion, on the other hand, is the fusing together of nuclei of light atoms to from heavier nuclei. This process is favourable for the very light atoms. The process releases an enormous amount of energy. In both fission and fusion, the energy liberated is equivalent to the loss of mass, which accompanies the reactions. Fusion produces much more energy per unit mass of reacting atoms than in fission.

For example, the fusion between two isotopes of hydrogen result in the formation of He.

$$ {}_{1}^{2}H + {}_{1}^{3}H \longrightarrow {}_{2}^{4}He + {}_{0}^{1}n + energy $$

Note: ${}_{1}^{2}H$ and ${}_{1}^{3}H$ are isotopes of hydrogen called deuterium and tritium, respectively.

Elements are synthesised in the stars by the process of nuclear fusion. For example, oxygen–16 is synthesised by the fusion of carbon–12 and helium–4. Gamma rays are emitted in the process.

$$ {}_{6}^{12}C + {}_{2}^{4}He \longrightarrow {}_{8}^{16}O + \gamma $$

Applications of Radioactivity

The following are some of the uses of radioisotopes:

(i) radioisotopes are used to study reaction mechanism

(ii) they are apply in the determination of molecular structure

(iii) they are used in agriculture for insect control and seed preservation

(iv) they are used for determination of age of artefacts using carbon dating

(v) they are used industrially for location of leakages and direction of flow

(vi) radioisotopes are used in medicine for treatment of cancer or tumour and goitre

(vii) they are used in inorganic chemistry e.g. for determination of solubility of sparingly soluble salts.

Questions

1.1 What are the three fundamental particles of an atom? Compare their relative masses and relative charges.

1.2 What are the mass number (A) and atomic number (Z) of the following nuclides?

(i) ${}_{8}^{15}O$ (ii) ${}_{13}^{27}Al$

1.3 What is meant by isotopic abundance?

1.4 The atomic mass of an element is 38.45 amu. It is made up of two isotopes. If one isotope has a mass and an abundance of 34.07 amu and 95.5%, respectively, what is the mass of the second isotope?

1.5 Describe briefly the principle of a mass spectrophotometry.

1.6 What is ionisation energy? Outline how ionisation energy can be determined experimentally from atomic spectra. Use hydrogen as an example.

1.7 How many unpaired electrons are present in one atom of (i) fluorine (ii) silicon (iii) arsenic (iv) phosphorus?

1.8 Write the electronic configuration for the following (i) Al^{3+} (ii) Mg^{2+} (iii) Cl (iv) S^{2-} (v) K.

1.9 The half-life of silver–112 is 3.2 hours. (a) Calculate the decay constant of silver–112. (b) How long will it take for its activity to decrease to 25% of the original activity? (c) How much of 0.1 g sample of silver–112 will remain after one day?

1.10 Identify the unknown (A – E) in the following equation or processes:

(i) ${}^{15}_{8}O \longrightarrow {}^{15}_{7}N + A$ (ii) ${}^{24}_{12}Mg + {}^{4}_{2}He \longrightarrow {}^{27}_{14}Si + B$ (iii) ${}^{238}_{92}U \longrightarrow {}^{234}_{92}Th + C$

(iv) ${}^{14}_{7}N + {}^{2}_{1}H \longrightarrow {}^{16}_{9}F + D$ (v) ${}^{238}_{92}U \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 5E$

1.11 Identify the nuclei ${}^{x}_{y}E$ in each of the following processes:

(i) ${}^{234}_{90}Th \longrightarrow {}^{x}_{y}E + {}^{0}_{-1}e$ (ii) ${}^{x}_{y}E + {}^{4}_{2}He \longrightarrow {}^{17}_{8}O + {}^{1}_{1}H$

1.12 The following is a part of the uranium decay series ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th \rightarrow {}^{234}_{91}Po \rightarrow X$. (i) What particles are emitted in each of the first two stages? (ii) If a beta particle is emitted in stage three, indicate the isotope X. (iii) If the activity of ${}^{234}_{90}Th$ is reduced to 75% in 48 days, what is the half-life of ${}^{234}_{90}Th$?

1.13 What do you understand by radioactivity? Use two equations to illustrate two types of radioactive decay. Explain five uses of radioisotopes.

1.14 (a) The approximate wavelength range of the visible spectrum is 400 nm to 700 nm. Express this range in Hz. (b) The wavelength of AM radio is about 250 nm, what is the frequency of this band?

1.15 Sodium vapour lamps, used for public lighting, emits 589 nm yellow light. How much energy is emitted by (a) excited sodium atom when it generates a photon? (b) A 1.0 mole of excited sodium at this wavelength

1.16 Using the Rydberg formula for hydrogen, calculate the wavelength and wavenumber for the transition from $n = 2$.

1.17 What is the energy of the photon emitted from a hydrogen atom when an electron makes a transition from $n = 7$ to $n = 2$? What is the wavelength of this photon?

1.18 How many subshells are there for $n =$ (a) 2, and (b) 3? What are the permitted values of l when n=3?

1.19 How many electrons can have the following number in an atom:

(a) $n = 2, l = 1$ (b) $n = 2; m = -2$ (c) $n = 2$ (d) $n = 3, l = 2, m = +1$?

1.20 Identify with reasons the species with larger radius in each of the following pairs:

(a) Cl and S (b) Cl^- and S^{2-} (c) Na and Mg (d) Mg^{2+} and Al^{3+}

1.21 Predict which atom of each pair has the greatest first ionisation energy (pay attention for any exception to the general trend): (a) Na or Mg (b) C or N (c) P or S.

1.22 Select the atom that has the greater affinity in the following pairs:

(a) S or Cl (b) C or O (c) Cl or Br (d) Si or P (e) N or O.